

Kamaljit Aulakh

315-551-5752 | 98kamaljitkaur@gmail.com | Portfolio | Github

PROFESSIONAL SUMMARY

Backend & Platform Engineer with experience building scalable **microservices**, automated **CI/CD** pipelines, distributed backend systems, and developer productivity tooling. Strong foundation in **Java**, **Python**, container orchestration, backend architecture, and test automation at scale. Known for reducing operational bottlenecks, improving system reliability, and delivering high-impact engineering solutions across fintech, enterprise integrations, and platform teams.

EXPERIENCE

Software Engineer II, Vertex Inc - Pennsylvania, USA

June 2025 – Current

- Designed and implemented a fault-tolerant **RESTful API** for an enterprise tax-content management system using **Python (FastAPI)**, **PostgreSQL**, **Redis**, and **Pydantic**, optimizing tax-ID record retrieval.
- Accelerated CI pipelines by 50% by migrating legacy Jenkins groovy based workflows to **GitHub Actions**, **Docker**, and **Azure Container Registry**, leveraging parallelized PyTest execution and matrix workflows.
- Built an AI-augmented failure-analysis workflow using **Microsoft Copilot agents** to automatically extract root causes from CSV/PDF artifacts, reducing triage time from 1 full day to <2 hours.
- Developing **Datadog** dashboards to monitor GitHub Actions pipelines and custom metrics (duration, failure rates, flaky tests), enabling teams to proactively optimize CI reliability and performance.

Web and Data Applications Developer, Open Source Program Office - New York, USA

October 2024 – May 2025

- Developed an open-source research workflow **plugin (PHP/WordPress)** used by 200+ researchers, replacing brittle legacy processes and enforcing **accessibility (WAVE)**, significantly improving data-exploration efficiency.
- Built a **JavaScript-based** inter-repository data transfer system bridging **HEAL and QDR** (Qualitative Data Repository), facilitating secure research data movement and validation; solution adopted and adapted by **Harvard**, demonstrating real-world impact.

Global Integrations Team Summer Intern, Capri Holdings (Michael Kors) - New Jersey, USA

June 2024 – August 2024

- Developed a secure internal audit-monitoring platform using **Node.js (Express)** and **Azure Tables** with **RBAC**, **JWTs**, and refresh-token rotation.
- Streamlined a CDC pipeline by integrating **Debezium SQL connector** into **Spring Boot** with custom **JOLT JSON** validation and transformation, enabling clean migration from legacy tooling to an event-driven architecture.

Software Engineer, IDFC FIRST Bank Ltd, Bangalore, India

June 2022 – August 2023

- Developed high-performance **RESTful** services using **Java Micronaut** and **Python Flask** for mobile and web banking, contributing to a 10% increase in online zero-balance account openings.
- Led a team of 4 engineers to build a **React.js admin panel** with responsive, reusable components, visualizing **Aerospike** data and generating analytical reports via **React Google Charts**.
- Implemented TDD (**JUnit**) achieving 90% coverage, integrated **SonarQube** for code quality, and optimized system performance with **JMeter**, monitored via **Prometheus + Grafana dashboards**.
- Containerized microservices with **Docker + Helm** and deployed on **Kubernetes** through **GoCD** CI/CD pipelines, achieving 98% uptime in production.

Systems Engineer, Tata Consultancy Services, Thane, India

Oct 2021 – June 2022

- Worked in an agile team writing backend logic across 10+ **microservices** using J2EE, **Spring MVC**, and **Spring Boot**, enhancing scalability and performance.
- Implemented front end features with **ReactJS**, **Redux** to integrate security enhancements (behavioral biometrics solution - BioCatch), leading to a 37% improvement in bot detection accuracy and reducing false positives by 20%.

EDUCATION

Syracuse University, College of Engineering and Computer Science, Syracuse, NY

August 2023 – May 2025

Master of Science in Computer Science

GPA: 3.90

Data Structures and Algorithms, Computer Security, Object Oriented Design

Mumbai University, Ramrao Adik Institute of Technology, Mumbai, India

August 2017 – June 2021

Bachelor of Engineering in Information Technology

GPA: 3.74

Web application development, Android Programming, Database Management Systems

SKILLS

Programming & Frameworks: Java, Python, JavaScript, Node.js, FastAPI, Flask, Micronaut, Spring Boot, React.js

Databases & Caching: PostgreSQL, Aerospike, Redis, MongoDB, SQL

Cloud & DevOps: Docker, Kubernetes, Helm Charts, GoCD, GitHub Actions, Azure, CI/CD pipelines

Monitoring & Testing: Prometheus, Grafana, Datadog, JMeter, PyTest, JUnit, SonarQube

Tools: Unix/Linux, Git, Postman, Jira, Confluence, Elasticsearch